



Electrochemical oxidation of solid Li_2O_2 in non-aqueous electrolyte using peroxide complexing additives for lithium–air batteries

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ABSTRACT

Using the anion receptor tris(pentafluorophenyl) borane as an additive to non-aqueous electrolytes, the solubility of solid Li_2O_2 can be dramatically increased through the Lewis acid–base interaction between boron and peroxide. The complexed boron–peroxide ions can be electrochemically oxidized with much better kinetics than the oxidation of solid Li_2O_2 on a carbon powder microelectrode. This discovery could lead to a new avenue for the development of high capacity, high rate, rechargeable, Li–Air batteries.

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1. Introduction

O_2 reduction products are mostly insoluble in non-aqueous electrolytes. They precipitate on the surface of the porous carbon electrode [1–3]. Re-oxidation of solid Li_2O_2 is a kinetically slow reaction. Yang et al. [4–8] reported on the synthesis and characterization of a group of boron-based anion receptors (BBAR). When used as additives in non-aqueous electrolytes, the Lewis acid center on boron in these anion receptors can complex with the anion (Lewis base centers) making some insoluble inorganic compounds (e.g. LiF , Li_2O_2 etc.) soluble. Due to the strong interaction between the anions in those inorganic compounds and the boron centers, their solubility in non-aqueous electrolytes can be increased by several orders of magnitude, when some BBARs such as tris(pentafluorophenyl) borane (TPFPB) are used. Such a boron-based Lewis acid additive has great potential to increase the solubility of the solid Li_2O_2 formed during the reduction of O_2 in non-aqueous electrolytes, thus keeping more surface of the porous carbon electrode active and in turn increase the capacity and discharge rate. Wu et al. [9] reported on the positive effects of using TPFPB in dissolving the Li_2O_2 and Li_2O in their Li–air cells. So far, no studies on the effects of TPFPB on the oxidation of solid Li_2O_2 in non-aqueous electrolyte have been reported in the literature. Using a powder microelectrode, we report here for the first time that the soluble boron–peroxide complex can be electrochemically oxidized. Such finding opens up a new avenue for the development of high capacity, high rate, rechargeable, Li–Air batteries.

The size of the microelectrode is substantially smaller than the thickness of the diffusion layer for the time scale of an electrochemical experiment, the current becomes time independent [10]. Therefore, a steady state of mass transport can be established more quickly, and the polarization homogeneity is higher than in its large electrode counterpart. The mass transport on a microelectrode with a smaller diameter is equivalent to that on a RDE with a higher rotation rate. However, unlike a rotating disk electrode (RDE), a powder microelectrode can be made of almost any solid powder material [11,12] while the RDE cannot. Thus, the microelectrode can be made with exactly the same carbon powder material as used in the gas diffusion electrode, making the experimental results closer to those obtained under real operating conditions. In addition, because there are no moving parts for a microelectrode, it is particularly suitable for the investigation of air sensitive systems such as a Li–air system, since the microelectrode can be accommodated easily in an air-tight electrochemical cell.

2. Experimental methods

All reagents were purchased from Sigma-Aldrich except tris(pentafluorophenyl) borane which was synthesized in-house at Brookhaven National Laboratory according to the procedures reported in the literature [4]. The moisture in electrolyte was less than 20 ppm as measured by Karl Fisher titration. The high surface area carbon material was supplied by Norit.

The detail description regarding the preparation and the construction of a powder microelectrode was reported in our earlier publication [13]. The electrode was made by sealing a 0.05 mm Pt wire in a soft, glass tubing. The electrode surface was first ground planar with

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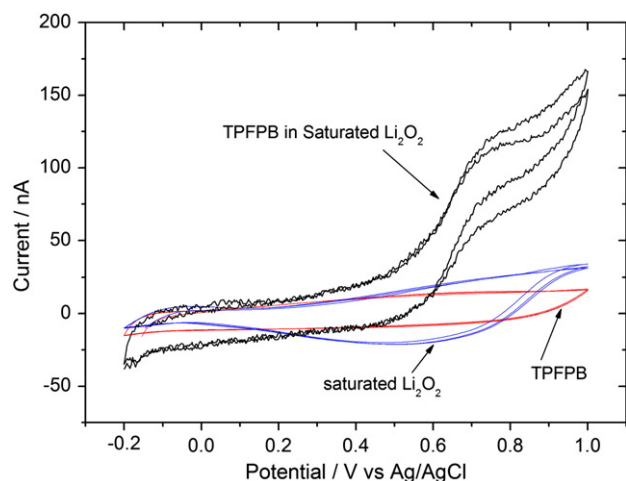


Fig. 1. Comparison of the cyclic voltammetry for the powder microelectrode of the same carbon material in Li_2O_2 saturated PC/ACN electrolyte and supporting electrolyte (0.5 M LiPF_6), PC/ACN electrolyte containing only TPFPB and supporting electrolyte (0.5 M LiPF_6) and Li_2O_2 saturated PC/ACN solution with 0.5 wt% TPFPB. Scan rate: 1 mV/sec.

coarse sand paper, and then finished to a mirror-like surface with micro-grained sandpaper and Al_2O_3 paste. The cavity then was formed by immersing the microelectrode in hot Aqua Regia for 10 minutes. The electrode was then washed thoroughly in an ultrasonic bath. The carbon powder was added into the cavity using the electrode as a pestle on a flat glass plate. The powder microelectrode was then sealed in a 25 ml 4-neck round bottom flask with a home-made Ag/AgCl reference electrode in the glove box with an O_2 and moisture concentration of less than 5 ppm. During the electrochemical experiment, a small positive pressure was maintained in the electrochemical cell with ultra-pure Ar gas. The positive pressure was established through the height of silicon oil in a bubbler, which also prevented the backflow of gases. The cell was housed in a Faraday cage during the measurement.

3. Results and discussion

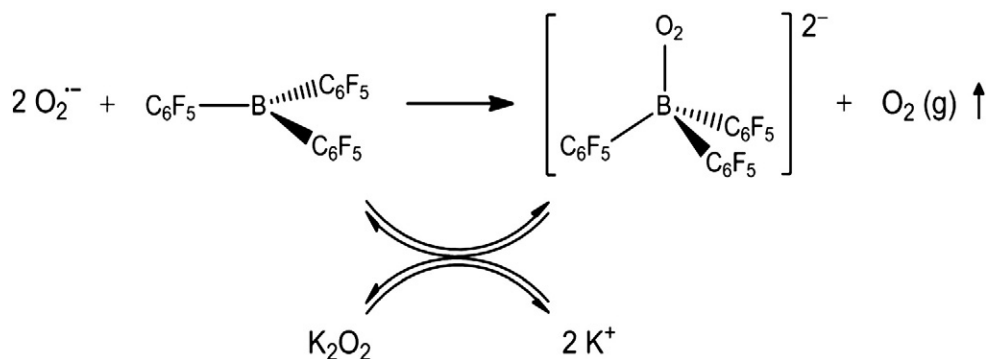
The solubility of Li_2O_2 in PC/ACN (1:5 volume ratio) was found to be 0.0010 M and 0.0268 M without and with the addition of 0.5% TPFPB (0.0391 M), respectively by ICP-OEX. The PC/ACN based electrolyte was found to be adequate for the Li–air system since the addition of PC helps to prevent any chemical reaction between the ACN and metallic Li [14]. It is clear that the solubility of Li_2O_2 has increased dramatically, by approximately 27 times, when TPFPB was added. The Lewis acid center can form a strong complex with the peroxide (O_2^{2-}) ion in Li_2O_2 . It has been reported that an electrolyte

made from dissolved Li_2O_2 with TPFPB has good conductivity. However, in order to use a TPFPB-type of Lewis acid additive in rechargeable Li–air batteries, the complexed peroxide ions should be electrochemically oxidized and its kinetics ought to be better in comparison with the oxidation of the solid Li_2O_2 .

It is worth to pointing out that electrochemical reduction of molecular O_2 is known to form superoxide ions (O_2^-), which can react with organic solvents and specifically carbonate solvents [15]. It has been discovered that TPFPB can also catalyze much faster disproportionation of O_2^- than the reaction between O_2^- and PC, which alleviates the issue of its reaction with PC and makes PC containing electrolyte viable for Li–air cell [16].

Fig. 1 shows the cyclic-voltammetry for the powder microelectrode in three different electrolytes: (1) Li_2O_2 saturated PC/ACN electrolyte; (2) PC/ACN electrolyte (0.5 M LiPF_6 as supporting salt) containing TPFPB only; and (3) Li_2O_2 saturated PC/ACN solution with 0.5 wt% TPFPB. The non-catalyzed carbon was the same material used in the gas diffusion electrode (GDE) for the Li–air battery studies. Due to the extremely low solubility of Li_2O_2 , there is a low quantity of peroxide ions in the electrolyte, and so no oxidation current can be measured on the carbon powder microelectrode in the saturated Li_2O_2 electrolyte. However, a substantial oxidation limiting current was observed on the same type of powder microelectrode when 0.5 wt% of TPFPB was added in the Li_2O_2 saturated solution. In order to determine whether the oxidation current was not due to a possible reaction of the TPFPB, the same experiment was carried out using a PC/ACN electrolyte containing only TPFPB and 0.5 M LiPF_6 . No oxidation current was observed, thus ruling out any reaction of the TPFPB. The limiting oxidation current clearly demonstrates that for the electrolyte with both Li_2O_2 and the TPFPB additive, the active peroxide ions were in the form of a soluble species. This demonstrates for the first time the oxidation of solid Li_2O_2 through a soluble peroxide complex species. The reaction is illustrated in Scheme 1. It is worth mentioning that electrochemical oxidation of electrolytes and carbon electrode itself could occur as side reactions during the recharge process of a Li–air battery, especially in the presence of catalyst [17]. In the current study, non-catalyzed carbon material was used and the side reactions were not evident as shown in Fig. 1.

These findings are quite important for the development of rechargeable Li–air batteries, because it provides a more flexible and efficient pathway for the oxidation of solid Li_2O_2 . The interfacial reaction of the soluble species on the surface of a porous electrode is more effective than that of the solid material, since the intrinsic resistance of the carbon electrode is increased when the porous electrode is covered with non-conductive Li_2O_2 and the dissolution of Li_2O_2 can suppress the reaction. The catalyst will be more effective for the oxidation of the soluble peroxide complex than for the solid Li_2O_2 . In the latter case, the catalyst can only be effective on the layer of Li_2O_2 , which has direct physical contact with the catalyst,



Scheme 1. The mechanism for the reduction of boron-peroxide complex assisted by the “shuttling” effect of TPFPB.

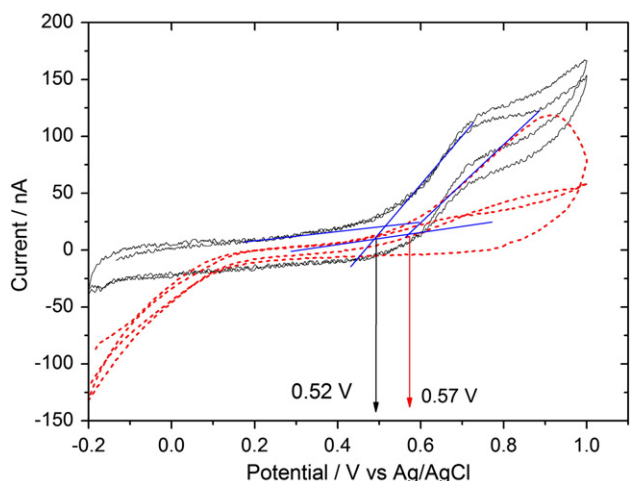


Fig. 2. Comparison of the cyclic voltammetry for the carbon powder micro-electrode in Li_2O_2 saturated PC/ACN solution with 0.5 wt% TPFPB (solid line) and 25 wt% solid Li_2O_2 mixed with 75 wt% carbon powder micro-electrode in 0.5 M LiPF_6 PC/ACN electrolyte (dash line).

but soluble species can easily access the entire catalyst surface. Such benefits are demonstrated in Fig. 2, which shows the comparison of the oxidation reactions for the soluble peroxide complex in the electrolyte and for the solid Li_2O_2 mixed with the carbon material. Powdered Li_2O_2 was mixed with the carbon material at a composition of 25% in the glove box after vacuum drying. The resulting mixture was loaded into the cavity of a microelectrode for electrochemical testing. It is apparent that both solid Li_2O_2 and soluble boron-peroxide complex can be oxidized electrochemically. However, a few distinguished differences can be observed. Although oxidation of solid Li_2O_2 occurs during electrochemical recharge, the soluble peroxide complex becomes oxidized at a lower potential. As shown in Fig. 2, the soluble peroxide complex starts to be oxidized at -0.52 V vs. Ag/AgCl while the oxidation of the solid Li_2O_2 starts at -0.57 V. The over-potential for the re-oxidation of the soluble peroxide complex is lower than that of solid Li_2O_2 , thus the soluble peroxide complex can be “recharged” at a lower potential. The kinetics for the oxidation reactions can be represented through the expression dI/dV . As shown in Fig. 2, the oxidation of the soluble peroxide complex has faster kinetics than that of the solid Li_2O_2 . The faster kinetics mean a higher rate capability during the recharging process. Interestingly, the oxidation current for the solid Li_2O_2 electrode decreases as its cycles, but the current remains unchanged for the oxidation of soluble peroxide complex. It is understandable that when the solid Li_2O_2 in contact with the carbon becomes oxidized in the microelectrode, the amount of reactant decreases as the cycling proceeds and so does the oxidation current. However, the concentration of soluble peroxide complex remains almost constant and therefore keeps the oxidation current stable during cycling. It would seem that the TPFPB acts like a carrier for the peroxide ions, and with the consumption of complexed peroxide ions through oxidation, the free TPFPB will complex more peroxide ions from the solid Li_2O_2 and bring them to the electrode to continue the oxidation process.

It is interesting to observe a substantial reduction current in the cyclic-voltammetry for the electrode prepared with a mix of solid

Li_2O_2 and carbon. Such a reduction current did not show in the cyclic-voltammetry for the carbon electrode in the electrolyte with TPFPB. Apparently, the O_2 formed during the oxidation of solid Li_2O_2 was trapped within the porous electrode and became reduced in the subsequent reduction process, while the O_2 generated through the oxidation of soluble additive-peroxide complexes were on the surface of the electrode and diffused away.

It is worth pointing out that as a Lewis acid, TPFPB could also form a complex with other Lewis bases e.g. anions in the electrolyte. It seems that the complex between TPFPB and peroxide ions is thermodynamically stable enough and the majority of TPFPB is complexed with peroxide ions. The results will be published later.

4. Conclusions

The results reported here show that TPFPB can form a stable complex with peroxide, which dramatically increases the solubility of solid Li_2O_2 . The TPFPB-peroxide complex can also be electrochemically oxidized on a porous carbon electrode. In comparison with the oxidation of solid Li_2O_2 , the over-potential for the oxidation reaction decreases and the kinetics improve. The presented results reveal a new avenue for recharging a Li- O_2 cell through the oxidation of the soluble peroxide complex instead of solid Li_2O_2 .

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